



Tribhuvan University
Institute of Science and Technology

Final Year Project Report

On

DosroGhar – A room rental system

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Submitted to:

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***In partial fulfillment of requirement for Bachelor's Degree in Computer
Science and Information Technology (B.Sc. CSIT)***



Tribhuvan University
Institute of Science and Technology
Department of Computer Science and Information Technology

STUDENT'S DECLARATION

We hereby declare that project report entitled “**DosroGhar - A room rental system**” submitted in the partial fulfillment of the requirement for Bachelor’s Degree in Computer Science and Informational Technology of Tribhuvan University, is our original work and not submitted for the award of any other degree, diploma, fellowship, or any other similar title or prize.

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by **Aditya Kc, Bashishta Raj Bhatta** and **Prasanna Malla** entitled “**DosroGhar - A room rental system**” in partial fulfillment of the requirements for the degree of Bachelor's Degree in Computer Science and Information Technology be processed for the evaluation.

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LETTER OF APPROVAL

This is to certify that this project prepared by **Aditya Kc, Bashishta Raj Bhatta and Prasanna Malla** entitled “**DosroGhar - A room rental system**” it is appreciable for the scope and quality as a project work in the partial fulfillment of the requirement for the degree of Bachelor’s in Computer Science and Information Technology.

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ACKNOWLEDGMENT

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ABSTRACT

“DosroGhar – A room rental system” is a comprehensive platform that facilitates seamless communication and interaction between property owners and tenants. Developed using Django, a powerful web framework for Python, DosroGhar offers a user-friendly interface for property owners to list their available rooms and manage their property details. Tenants can easily explore room listings, view property details, and connect with property owners. The platform includes user authentication, custom profiles for both owners and tenants, and features like uploading photos for property listings. DosroGhar aims to simplify the process of finding and listing rental accommodations, creating a convenient and efficient solution for both property owners and tenants.

Keywords: *User Authentication, Design, Listing, Interaction, Leaflet*

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LIST OF ABBREVIATIONS

HTML	Hyper-Text Markup Language
DFD	Data Flow Diagram
ER	Entity Relationship
UI	User Interface
CSS	Cascading Styling Sheet
JS	JavaScript
DB	Database
QA	Quality Assurance
SRS	System Requirement Specification

CHAPTER 1: INTRODUCTION

1.1 Introduction

In the academic journey of final-year students, the successful completion of a significant project represents a defining moment. However, this moment comes with its fair share of challenges. First and foremost, students must embark on the daunting task of identifying a suitable project idea. It's not merely about selecting any idea; it must align with their area of study, be engaging, and have a clear purpose.

Once an idea is chosen, the next hurdle involves transforming it into a practical project. This requires careful planning, extensive research, and problem-solving skills. Additionally, students need to manage their time effectively, consider available resources, and anticipate potential obstacles that may arise during the project's execution. In essence, while this final-year project is a milestone that showcases their learning and growth, the journey to reach it demands considerable effort and strategic thinking.

So, to fill this gap, our group has decided to make a platform named as “DosroGhar - A room rental system”. DosroGhar is a service that is designed to solve the problem that arises during searching a room, flat, house, apartment, and land for rental in respect to their budget. DosroGhar is mainly focused on Room, Flat, House, Apartment for rent.

An online room rental service is a digital platform or website that connects property owners (hosts) with individuals seeking accommodations (guests) for short-term stays.

In the context of Nepal, searching for a suitable room in urban and college areas is a challenging task. To address this challenge, a digital solution called "DosroGhar" has been introduced. Through DosroGhar, house owners and room renters can search and find rooms according to their temporary residence needs.

1.2 Problem Statement

In Nepal, finding accommodation in city, college, or industrial areas can be quite challenging due to its status as a developing country with diverse cultures. This challenge extends to both urban and rural areas with high population densities. Additionally, there are several other problems associated with this system:

- **Tedious Searching:** Tenants face difficulties in finding suitable rental properties due to the lack of a centralized platform, leading to time-consuming searches across multiple websites or physical visits to various properties.
- **Limited Visibility:** Property owners struggle to reach a broad audience of potential tenants.
- **Vague Location Details:** Lack of precise location information in property listings makes it challenging for tenants to determine proximity to amenities and transportation, leading to uncertainty and inconvenience during the search.

1.3 Objectives

Some of our project's objectives are as follow:

- To provide a platform where users can easily find accommodation for short term stays.
- To build a room rental system using the QuickSort algorithm to recommend rooms sorted from lowest to highest price.
- To provide exact location of room/flat using map.

1.4 Scope and limitations

1.4.1 Scope

Scope of an online room rental system includes facilitating users to easily search, view, and manage accommodations remotely through a digital platform. It allows property owners to list their rooms, apartments, or houses for rent, enabling them to reach a wider audience and maximize occupancy rates. Users can browse through various options, filter based on their preferences, view photos and descriptions, and make conversation. Additionally, the system may incorporate features such as recommending cheaper room to help users make informed

decisions. Overall, the scope encompasses streamlining the process of finding and renting accommodations efficiently and conveniently.

1.4.2 Limitations

- Not all properties may be listed, reducing options for renters.
- Requires stable internet connection for browsing.
- Lack of personal interaction compared to traditional bookings.
- The system is based on web only and there is no dedicated Android or iOS applications.

1.5 Development Methodology

Incremental Model is a process of software development where requirements divided into multiple standalone modules of the software development cycle. In this model the requirements are broken down into different standalone modules which work as individual increments. Each increment goes through the requirement analysis, design, coding and testing phases.

Phases of Incremental Model

The various Phases that are involved in Incremental Model include:

1. Requirement analysis

In the first phase of the incremental model, the product analysis expertise identifies the requirements. And the system functional requirements are understood by the requirement analysis team. To develop the software under the incremental model, this phase performs a crucial role.

2. Design

In this phase of the Incremental model of SDLC, the design of the system functionality and the development method are finished with success. When software develops new practicality, the incremental model uses style and development phase.

3. Coding

In this phase, coding is done based on the system design developed. A working product is developed by the completion of this phase.

4. Testing

In the incremental model, the testing phase checks the performance of each existing function as well as additional functionality. In the testing phase, the various methods are used to test the behavior of each task.

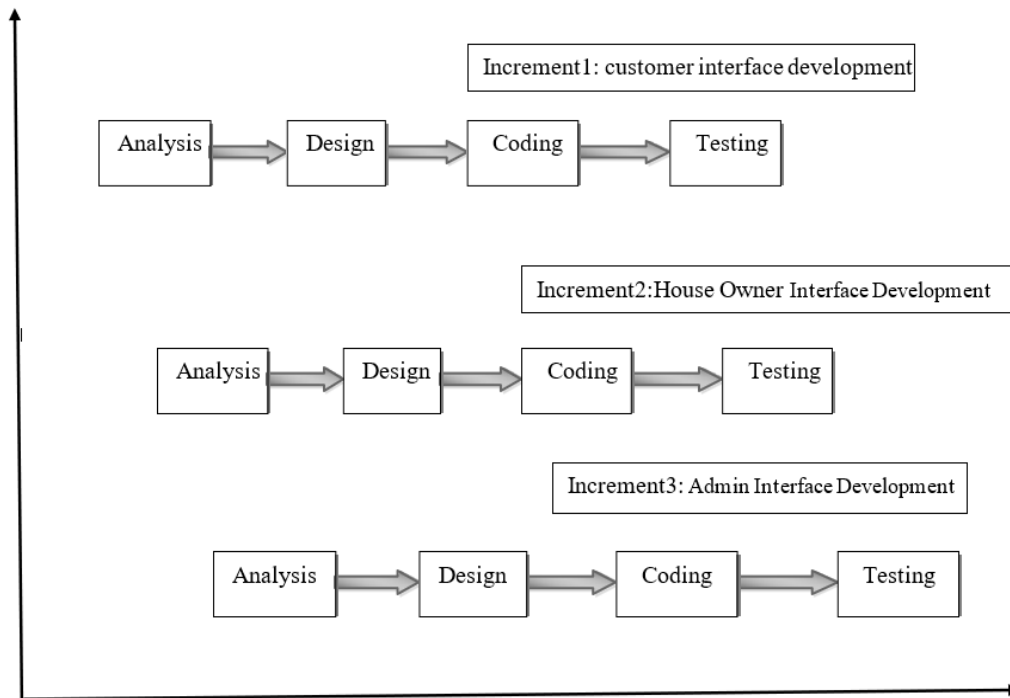


Fig 1.1: Increment Process Model

1.6 Report Organization

The report has been meticulously prepared in accordance with the guidelines set forth by Tribhuvan University. It is structured into distinct chapters, each containing a multitude of subchapters delineating its respective content. This organizational framework ensures clarity and coherence throughout the document, facilitating comprehensive understanding and navigation.

The report is divided into sections which include:

Table 1: Report Organization Table

Preliminary Section	• Title Page • Acknowledgement • Abstract • Table of Contents • List of Abbreviations • List of Figures • List of Tables
Introduction Section	• Introduction and Background • Problem Statement • Objectives • Scope and Limitations • Development Methodology • Report Organization
Background Study and Literature Review Section	• Background Study • Literature Review
System Design Section	• Design • Algorithm Details
Implementation and Testing Section	• Implementation • Testing • Result Analysis
Conclusion and Future Recommendation Section	• Conclusion • Future Recommendation

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Online room rental system is a service that is designed to solve the problem that arises during searching a room, flat, house, apartment for rental in respect to their budget. This is mainly focused on Room, Flat, House, Apartment for rent.

An online room rental service is a digital platform or website that connects property owners (hosts) with individuals seeking accommodations (guests/tenant) for short-term stays. These services have become increasingly popular in recent years, revolutionizing the way people find and book lodging.

In the context of Nepal, searching a suitable room in city area and in college area is challenging task. To overcome this, we brought a digital solution called “DosroGhar” where the house owner and room rental can search and find the room according to their needs for their temporary residence.

After the development of the Internet and communication technology, everything became digitalized and became most popular. After the COVID 19 pandemic online service became most popular. It aims to solve the upcoming problem that arises during searching for a new room for temporary lodging.

2.2 Literature Review

Looking at the current online room rental system, there are a handful of such systems that have been providing online services to tenants and property owners. So, in this section, two of such existing systems that resemble this project will be reviewed.

House is referred to as any house, story, unit, or room of the house or land or cartilage occupied by such a house. If a person who has the right, possession, or ownership in a house lends gives the house to another person to use it for a certain period of time, then it is called house rent [1].

House rent law in Nepal is governed by Muluki Civil Code 2074.

The 10% tax (in the case of Lalitpur and Kathmandu municipal authority) on house rental payment to natural persons should be deposited in respective offices. The land or land and building or building disposed at proceeds less than Rs 1,000,000 don't fall under the category of non-business chargeable assets (NBCA) [2].

Except as in the agreement, it is the duty of the house owner to pay the tax leviable by law for the house rented. If the house owner doesn't pay the tax, the house renter may pay the tax and such payment is deducted from the house renter while paying the rent.

According to the Ministry of Finance, it is the responsibility of the Committee to study the necessary studies to develop an appropriate revenue system by timely evaluation of revenue policy, tax policy, tax structure and working procedure. On the basis of the national economic policy, the committee has been formed to make a timely assessment of the revenue policy, tax related policy, tax structure and working procedure, including suggestions to the Government of Nepal [3]. The committee is responsible for the development of the tax policy with the responsibility of giving suggestions to the government for the necessary study research. The committee has been formed to review internal sources, expand tax net, tax exemption and double tax exemption agreement.

Some Existing System are:

- **GharBheti.com:** Gharbeti.com stands as one of the most prominent room rental systems in Nepal, renowned for its widespread popularity. Operating primarily in urban hubs like Kathmandu, Pokhara, and Butwal, it facilitates customers in booking rooms and flats for temporary residence. As part of its service, the platform charges a nominal broker fee for facilitating these transactions [4]. Gharbeti.com streamlines the booking process, offering users access to customer reviews for informed decision-making. However, it currently lacks features such as pre-booking options for available rooms and flats, as well as real-time communication capabilities between property owners and customers.
- **Rental Nepal:** Rental Nepal is one of the most popular room management systems in Nepal. It allows customer to pre-book the houses for temporary resident [5]. It allows customers to pre-book houses for short-term stays, providing a convenient solution for both property owners and tenants. However, the platform currently lacks real-time communication features between property owners and customers, which could enhance the booking experience. Additionally, while the platform offers basic functionality, there is room for improvement in terms of user interface design to make it more intuitive and user-

friendly. Despite these limitations, Rental Nepal remains a valuable resource for individuals seeking temporary housing solutions in Nepal.

- **Hamro Kotha Online Room Management System:** Through the official website of Hamro Kotha a customer can request a rooms, flats and apartment by filling the required information available at the contact form. The Admin of Hamro Kotha searches the room with respect to the customer's needs and requirements. After they found the suitable property Admin of Hamro Kotha informs the tenant and proceed with certain conditions [6]. Customer can search the room and select the rooms, flats, apartment as per their needs. Customer can request room booking through contact form given. After submitting the contact form the Admin of Hamro Kotha contacts with owner and provides update to the customer. If a customer is satisfied with the agreements provided with house owner, he/she can proceed booking by making certain amount payment as an advance to house owner. The Admin of Hamro Kotha takes certain charges from both House owner and customer. These are the features that are known to the customer and house owner. But the website fails to connect both host and customer. It does not provide transparency while proceeding booking of rooms, flats and apartment. It is also not specified what types of facilities he/she is getting from the room.

CHAPTER 3: SYSTEM ANALYSIS

3.1 System Analysis

Seeing the possibility of frequent changes that may occur during the development, the Incremental model was used to develop the web application. The detailed methodologies used to develop the application are described in the following subsections.

3.1.1 Requirement Analysis

3.1.1.1 Functional Requirement

The functional requirements of DosroGhar are as follows:

- User Registration: The system should allow owner/tenant to register their account.
- Add/View Room: Owner should be able to add their property with specified details and location using Leaflet map. Tenant should able to view and can send message to owner of room.

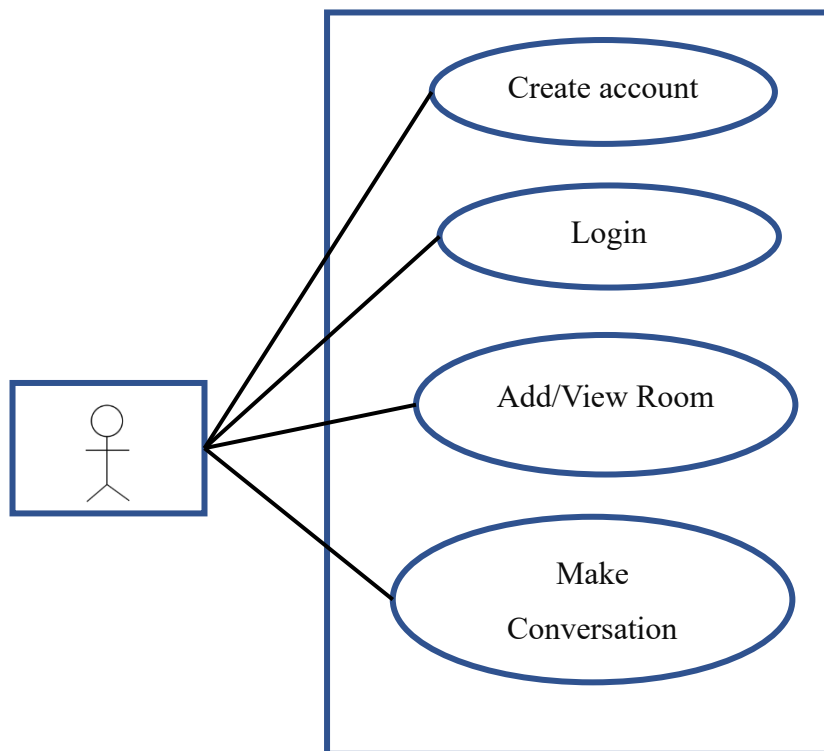


Fig 3.1: Use Case Diagram

3.1.1.2 Non-Functional Requirement

- Easy and Simple UI: The user interface of application is simple and user friendly.
- The system should be able to handle an unlimited number of users at a time.

3.1.2 Feasibility Analysis

A feasibility study, also known as feasibility analysis, is an analysis of the viability of an idea. It describes a preliminary study undertaken to determine and document a project's viability. The results of this analysis are used in making the decision whether to proceed with the project or not. In short, a feasibility analysis evaluates the project's potential for success, following feasibility analysis was performed prior to working on the project:

3.1.2.1 Technical feasibility

All the tools and software products required to construct DosroGhar is easily available in internet. It doesn't require special environment to execute.

3.1.2.2 Operational Feasibility

User interface of the system is simple and easy to navigate for user which ensure the smooth operation of system. In addition to this system is flexible to change, scalability and maintenance is easier to apply.

3.1.2.3 Economic Feasibility

There is no need of extra hardware components and paid software to build the system. The price of software needed to develop and operated is incurred only once. Its maintenance cost is cheaper as compared to other.

3.1.2.4 Schedule Feasibility

With available resources and platform, the project can be completed within the estimated time period. So, based on academic schedule, the system is practically feasible and achievable. The project expands over a period of 84 days with Task being divided along with duration in the time frame on basis of weeks as shown in below table. The planning phase of the project began on October 10, and the project concluded on January 15.

Table 3.1: Gantt Chart

Key features	Duration (in weeks)											
	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th
Planning												
Analysis and Design												
Coding												
Testing												
Evaluation												
Documentation												
Deployment												
Presentation												

3.1.3 Analysis

Project analysis is a complex process that can be used to improve the efficiency and effectiveness of projects. It helps identify and correct problems early in the project, so they can be fixed before they become major issues.

3.1.3.1 Data modeling using E-R Diagram

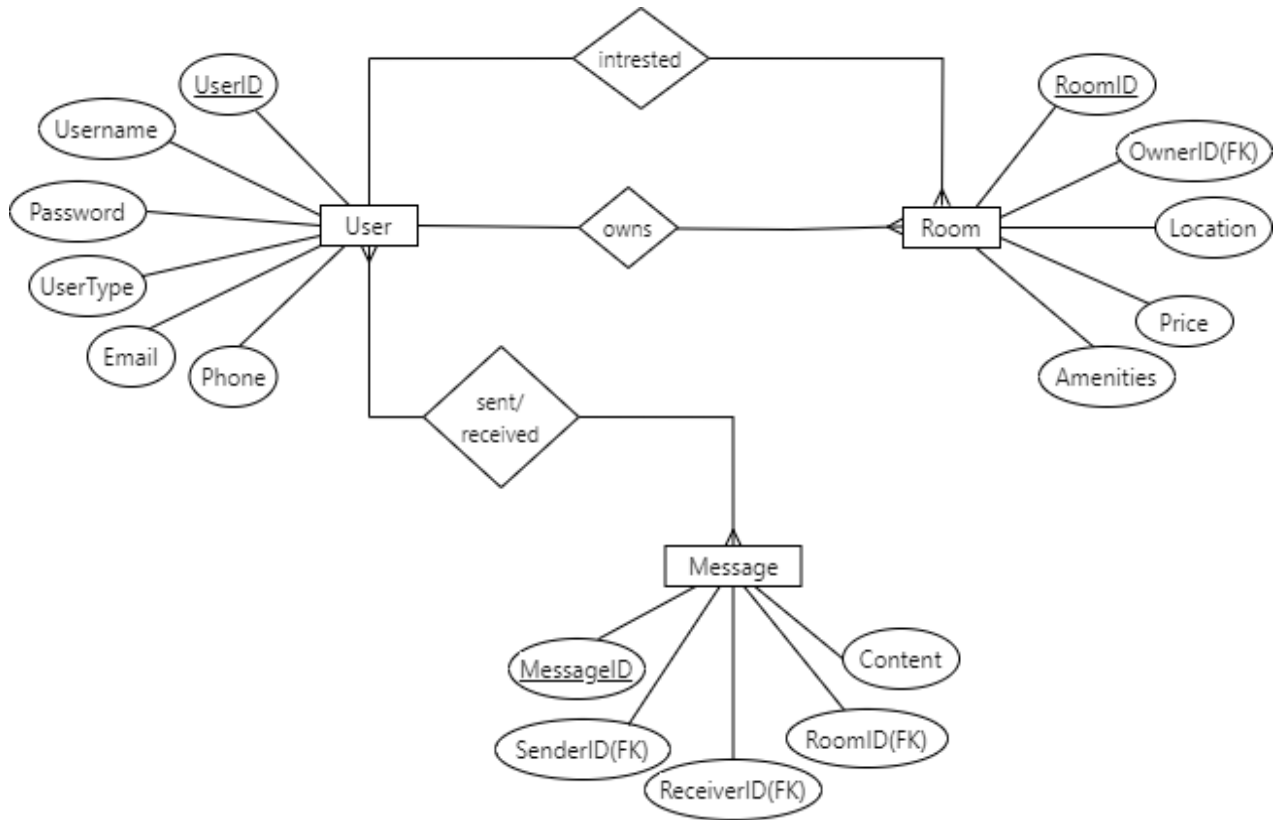


Fig 3.2: E-R Diagram

3.1.3.2 Level 0 DFD

The context level data flow diagram describes the whole system. The 0 level DFD describes all user modules who operate the system. The below DFD of the system shows the two users can operate the system.

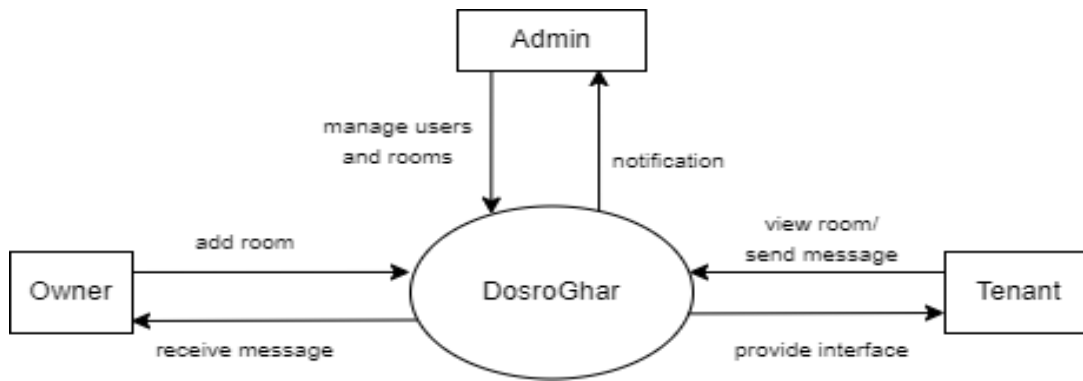


Fig 3.3: Level 0 Data Flow Diagram

3.1.3.3 Level 1 DFD

In Level 1 DFD the process of whole system is decomposed with multiple process. Necessary data source are introduced and system is represented as combination of multiple process, entities and data sources. This give more detailed picture of system and how individual components together to form a system and what process they each perform. This system has authentication process and project registration process represent by authentication system and project library system.

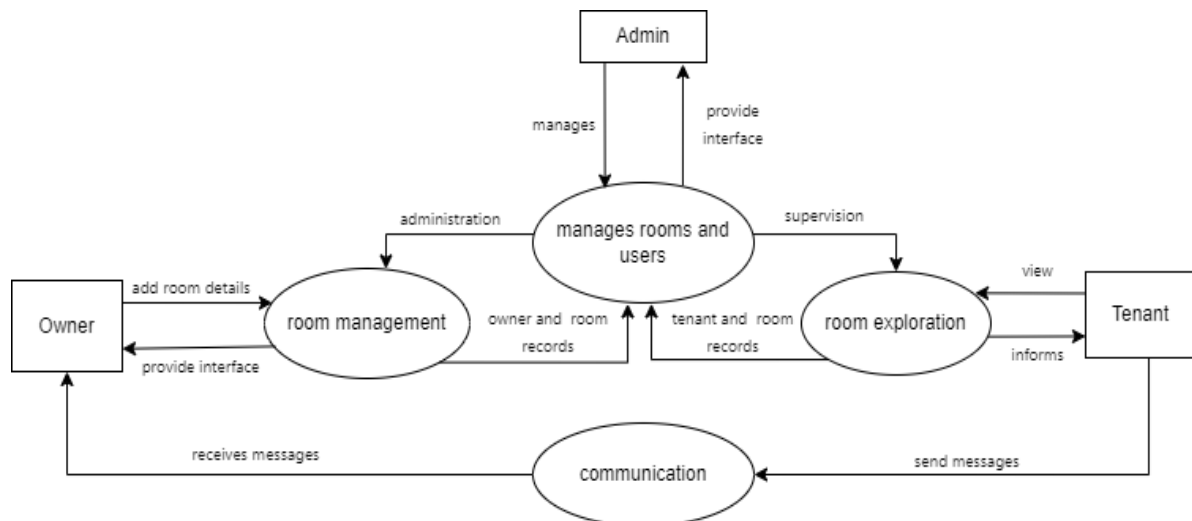


Fig 3.4: Level 1 Data Flow Diagram

CHAPTER 4: SYSTEM DESIGN

4.1 Design

System design refers to the process of defining the architecture, components, modules, interfaces, and data for a software system. It involves creating a blueprint or plan for the system that specifies how it will be built, how it will function, and how it will meet the requirements of the clients.

4.1.1 Database Design

The "DosroGhar" project's database design includes four tables: Admin, Owner Profile, Tenant Profile, and Room. Admin stores administrator details, Owner Profile contains property owner information, Tenant Profile holds tenant data, and Room represents rental rooms with attributes like price, location, and availability, facilitating platform functionality and data management.

Table 4.1: Admin Table

S.N	Attribute	Data Type	Constraints
1	_id	Integer	Primary Key
2	Password	Varchar	Not Null
3	Last_login	Date-Time	Not Null
4	Is_superuser	Boolean	Not Null
5	Username	Varchar	Not Null
6	Lastname	Varchar	Not Null
7	email	Varchar	Not Null
8	Is_Staff	Boolean	-
9	Is_Active	Boolean	-
10	Date_joined	Boolean	-
11	Firstname	Varchar	Not Null

Table 4.2: Owner Profile Table

S.N	Attribute	Data Type	Constraints
1	_id	Integer	Primary Key
2	FullName	Varchar	Not Null
3	Email	Varchar	Not Null
4	Password	Varchar	Not Null
5	Phone_number	Integer	Not Null
6	Citizenship_photo	Blob	Not Null

Table 4.3: Tenant Profile Table

S.N	Attribute	Data Type	Constraints
1	_id	String	Primary Key
2	Name	Varchar	Not Null
3	Email	Varchar	Not Null
4	Phone_number	Integer	-
5	Address	Varchar	-
6	Photo	Blob	Not Null

Table 4.4: Room Table

S.N	Attribute	Data Type	Constraints
1	_id	String	Primary Key
2	Tittle	Varchar	Not Null
3	No_of_room	Integer	Not Null
4	No_of_hall	Integer	Not Null
5	No_of_bathroom	Integer	Not Null
6	Location	Varchar	Not Null
7	Price	Decimal	Not Null
8	Room_photo	Blob	-
9	Hall_photo	Blob	-
10	Kitchen_photo	Blob	-
11	Bathroom_photo	Blob	-
12	Parking_area_photo	Blob	-
13	Owner_id	Integer	Not Null
14	Created_at	Date-Time	-
15	Description	Text	-
16	Updated_at	Date-Time	-
17	Latitude	Decimal	Not Null
18	Longitude	Decimal	Not Null

4.2 Algorithm Details

The QuickSort algorithm efficiently sorts room prices in the project. Room data is stored in a list, sorted by price. A recommendation function suggests rooms within a specified budget, integrated into the user interface. Testing ensures accuracy, providing users with efficient access to suitable rental accommodations.

4.2.1 Quicksort Algorithm

Quicksort is a highly efficient sorting algorithm that follows the divide-and-conquer paradigm. It works by selecting a 'pivot' element from the array and partitioning the other elements into two sub-arrays according to whether they are less than or greater than the pivot. The process is then applied recursively to the sub-arrays. The efficiency of quicksort is attributed to its in-place sorting nature and its ability to sort in $O(n \log n)$ time on average.

4.2.2 Pseudo-Code

```
def quicksort(arr):
    if len(arr) <= 1:
        return arr
    pivot = arr[len(arr) // 2]
    left = [x for x in arr if x.price < pivot.price]
    middle = [x for x in arr if x.price == pivot.price]
    right = [x for x in arr if x.price > pivot.price]
    return quicksort(left) + middle + quicksort(right)
```

CHAPTER 5: IMPLEMENTATION AND TESTING

5.1 Implementation

In the implementation phase of this project, several key technologies are utilized to build a robust and user-friendly system. Here's how each technology contributes to the development process:

5.1.1 Tools Used

5.1.1.1 Front End Tools Used:

- **HTML:**

HTML is the standard markup language for documents to be displayed in a web browser. HTML was used to display the content and structure of the webpages in the project.

- **CSS:**

CSS stands for Cascading Style Sheets. This is a very powerful language for formatting web pages and has been of great help in the project. It was used to format the user interface to make it more appealing to the users.

- **JavaScript:**

It is a scripting language of HTML and the web. JavaScript is used for enabling interactive webpages in our project.

5.1.1.2 Backend Tools Used:

- **Python:**

Python is often used to develop the back end of a website or application—the parts that a user doesn't see. Python's role in web development can include sending data to and from servers, processing data and communicating with databases, URL routing, and ensuring security.

5.1.1.3 Database Technology Used:

- **SQLite:**

SQLite is a lightweight, serverless, and self-contained relational database management system (RDBMS) renowned for its simplicity and efficiency. It is embedded directly into applications, eliminating the need for a separate server process. SQLite is often employed as a local database for mobile apps, embedded systems, and desktop applications due to its minimal setup requirements and small footprint.

5.1.2 Implementation Details of Modules

Modules are the partition of any project done to ease the task of development. Different modules are designed so that debugging and other development phases get the easiest implementation.

The different modules of the system are:

5.1.2.1 Register/Login page

This page allows user to register to the system if they haven't already or if they had then allows to login to the application.

5.1.2.2 Home page

This page is the first loading page of the system. It contains list of recommended house and apartment. Customer can view the details of respective recommendation listed there according to their interest.

5.1.2.3 Dashboard page

This page allows user to view their Dashboard by providing necessary details of the users provided in the system.

5.1.2.4 About us Page

This page introduce user with what the system is all about. How it came into existence, its objectives, future vision and introduce core team members of the system.

5.2 Testing

Testing plays a crucial role in ensuring the quality and reliability of software systems. It involves systematically evaluating the functionality, performance, and usability of the application.

5.2.1 Test Cases for Unit Testing

Unit-Testing is a type of software testing where individual units or components of a software are tested. The purpose is to validate that each unit of the software code performs as expected. The following test scenarios were used to test the system after the build is completed.

Table 5.1: Test Case for User Registration and Login

Test Id	Test Case	Input	Expected Output	Observed Output
1	Successful User registration	Username: prasanna1 Email: prasanna@gmail.com Password:ram@123	Successful registration	User is registered and logged in and go to home page.
2	User Login Failed	Username: prasanna1 Password: ram123	Login failed	User unable to login to the system
3	User login fail	Username: prasanna Password: ram@123	Login failed	User unable to login to the system

5.2.2 Test Cases for System Testing

System testing is defined as testing of a complete and fully integrated software product. System testing of software or hardware is testing conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. System testing falls within the scope of black box testing, and as such should require no knowledge of inner design of the code or logic. One of the types of system testing is the usability testing which is performed in the system.

Table 5.2: Test Case for System Usability Testing

Test Id	Test Case Description	Expected Output	Observed Output	Status
1	Click on various links on the system.	Link should take users to another web page according to the on-page URL.	Users were able to get the expected web pages after clicking the links.	Pass

5.3 Result Analysis

All the modules of the system are working and the output of the system is up to expectations. The developed system provides us with the features of searching the rooms and flats. Customer can get through the details and find out the choice of their interest.

This application uses Quicksort algorithms to sort the relevant and related rooms and flats to customer according to Price. Customer than can explore and get details of rooms and apartments of their similar interest and gain a clear picture of distinction of which one to consider and which not to through their exploration of multiple choices and suggestions provided by the system.

Main focus is given on creating the user-friendly interface so that customer and property owner come together in a common space fulfilling the gap and making a system that bridge to make the rental process smooth and efficient making it simple and handy.

Overall, the system is working according to expectations and without any problem.

CHAPTER 6: CONCLUSION AND FUTURE RECOMMENDATIONS

6.1 Conclusion

The system, built with HTML, JavaScript, Python, and SQLite, effectively fulfills its intended objectives. With optimized efficiency, it streamlines room/flat renting processes, addressing manual system limitations. Moreover, it alleviates staff burdens, shortens bus terminal reservation times, and significantly enhances overall operational efficiency.

6.2 Future Recommendation

Looking ahead, the system serves as a solid foundation for enhancing room rental processes through its integration of HTML, JavaScript, Python, and SQLite. To maintain its relevance and effectiveness, several future recommendations have been identified. These strategic enhancements aim to elevate the capabilities of the system, enhance user experience, and improve overall adaptability.

This system can be further improved by taking into consideration the following recommendations:

1. Integrating booking and payment options to enhance the user experience and streamline the rental process.
2. Adding a feature for house owners to share their experience and the behavior of customers acquiring rooms. Similarly, customers can rate and comment on their opinions regarding flats, rooms, and owner behavior.

REFERENCES

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APPENDICES

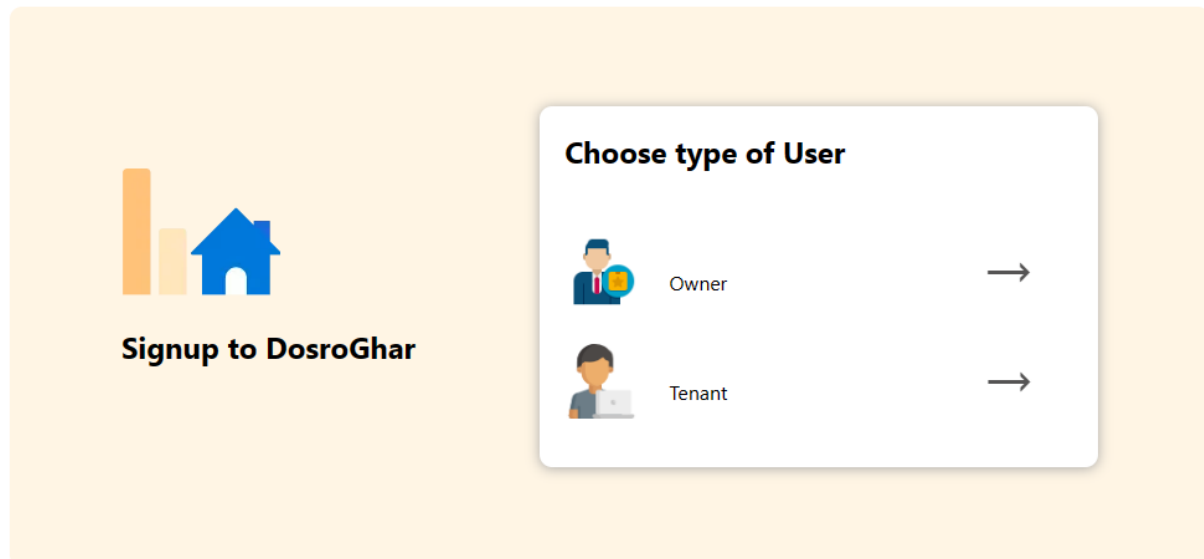


Fig 6.1: Signup Page

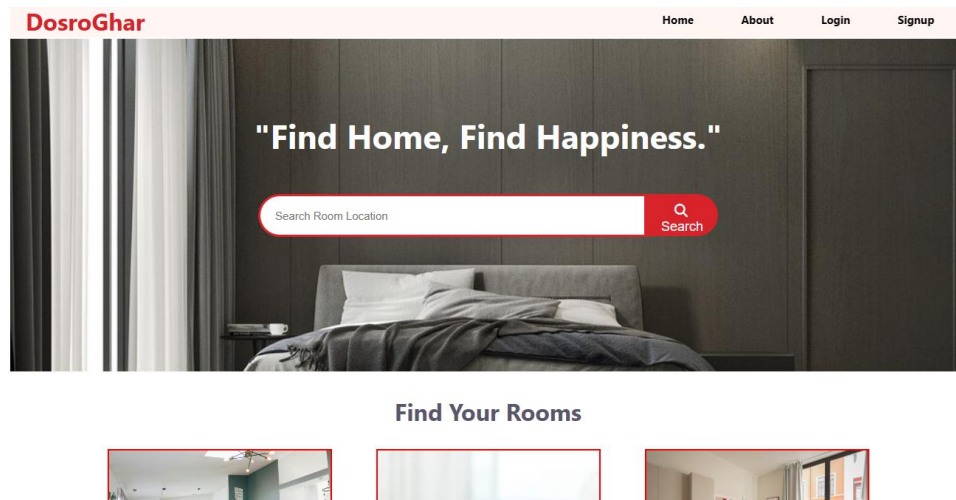


Fig 6.2: Home Page

DosroGhar

Tenant Dashboard

Name: Prasanna Malla

Address: Kailali

Phone Number: 9811686161

Your Photo



DosroGhar

Welcome to Your Dashboard, nishant.shakya
Your Profile:

Name: Nishant Shakya
Phone Number: 9999999999
Citizenship Photo:



Your Rooms:

Room for Csit Student Only - [View](#) [Delete](#)

[View Room Messages](#)

Room for BCA - [View](#) [Delete](#)

[View Room Messages](#)

Teacher Room - [View](#) [Delete](#)

[View Room Messages](#)

Fig 6.3: Dashboard Page of Tenant and Owner